



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2018 Preliminary Course Yearly Examination

12th of September, 2018

General instructions

- Working time – 2 hours 30 minutes (plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.

QUESTIONS 1-9

- Commence each new question on a **new page**. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ Mr Berry

☐ Ms Lee

☐ Mr Hwang

☐ Mr Lin

☐ Mr Ireland

☐ Ms Ziazaris

☐ Dr Jomaa

Marker's use only.

QUESTION	1	2	3	4	5	6	7	8	9	Total
MARKS	18	24	13	11	9	9	11	8	7	110

Start EACH question in a SEPARATE answer booklet.

Start a NEW booklet

Question 1 (18 marks)

Marks

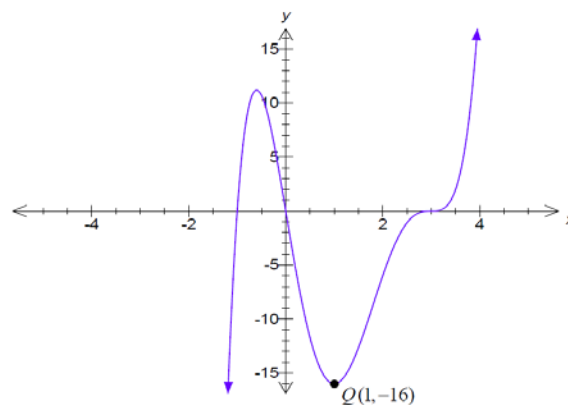
- a) Fully factorise $2x^3 - 128$ 2
- b) Solve $\frac{2x}{5-x} \geq 1$ 3
- c) Solve $|2x - 1| < 3x + 2$ 3
- d) Solve $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 6 = 0$ 3
- e) Given $\sin \theta = \frac{4}{5}$ and $\frac{\pi}{2} \leq \theta \leq \pi$ find $\sin 2\theta$ 2
- f) Simplify $\frac{\cos(270-\theta)}{\cos(180-\theta)}$ 2
- g) Solve $2\tan^3 \theta - 3\tan^2 \theta - 2\tan \theta + 3 = 0$ for $0^\circ \leq \theta \leq 360^\circ$, giving your answers to the nearest minute, where necessary 3

Start a NEW booklet

Question 2 (24 marks)

Marks

- a) Show that $\sin\left(x + \frac{\pi}{4}\right) = \frac{\sin x + \cos x}{\sqrt{2}}$. 4
Hence or otherwise solve $\frac{\sin x + \cos x}{\sqrt{2}} = \frac{\sqrt{3}}{2}$ for $0 \leq x \leq 360^\circ$
- b) Prove that $\frac{\cos 2\theta}{\sin \theta} + \frac{\sin 2\theta}{\cos \theta} = \operatorname{cosec} \theta$ 3
- c) Find the size of the acute angle between the lines $y = 2x + 3$ and $4x - y + 1 = 0$ 2
- d) The point $(8, -5)$ divides the interval AB externally in the ratio 3:2. If A is the point $(-1, 4)$, find the coordinates of $B(x, y)$. 2
- e) Sketch the region of the number plane where $y \leq 1$ and $y < |x - 2|$ hold simultaneously 3
- f) Write $\frac{2^{1001} + 2^{1000}}{3}$ as power of 2 2
- g) 4



- i) The above graph has equation $f(x) = ax(x - b)(x + c)^d$. Find the values of a, b, c, d 3
- ii) Write down the domain of $y = \sqrt{f(x)}$ 1

Start a NEW booklet

Question 3 (13 marks)

Marks

a) Differentiate and fully simplify

(i) $y = x(1 - x)^7$ 2

(ii) $y = x \cdot \sqrt[3]{x}$ 2

(iii) $y = \frac{5-x^2}{2+x}$ 2

b) Find the value of a given that the gradient of the tangent at $(2, 4)$ on $y = 3x^2 - 2ax + 1$ is -1 . 2

c) For what values of x is the curve $y = 4x - x^3$ concave upwards? 2

d) Differentiate from first principles $f(x) = 3x^2 - 2x$ 3

Start a NEW booklet

Question 4 (11 marks)

Marks

a) Consider the function $f(x) = \frac{x+1}{x^2+3}$ 6

i) Find the x and y intercepts

ii) Find the coordinates of any stationary points on the curve $y = f(x)$ and determine their nature

iii) Find $\lim_{x \rightarrow \infty} \frac{x+1}{x^2+3}$

iv) Sketch $y = f(x)$

b) If α and β are the roots of the equation $4x^2 - 2x - 1 = 0$ find (without solving for x) 5

i) $\frac{1}{2\alpha} + \frac{1}{2\beta}$

ii) $\alpha^2 + \beta^2$

iii) $|\alpha - \beta|$

Start a NEW booklet

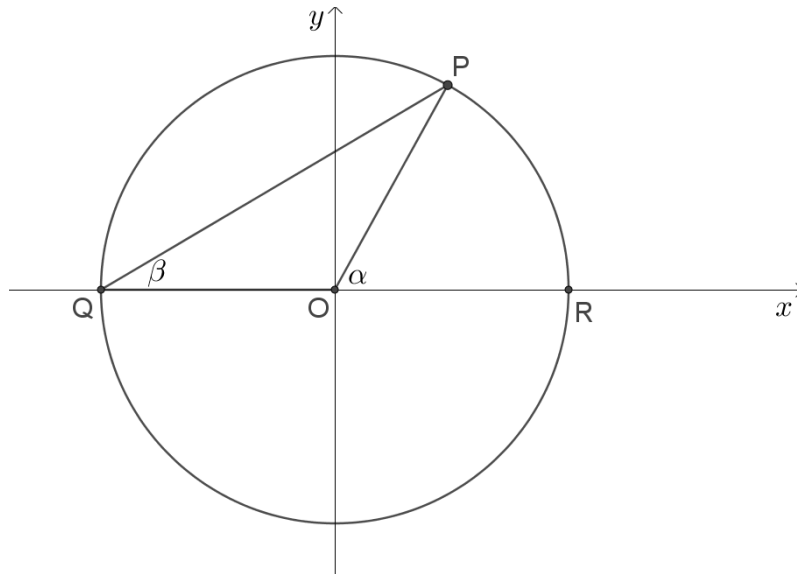
Question 5 (9 marks)

Marks

a) Find the values of k for which $(k + 1)x^2 - 2(k - 1)x + (2k - 5)$ is a perfect square 3

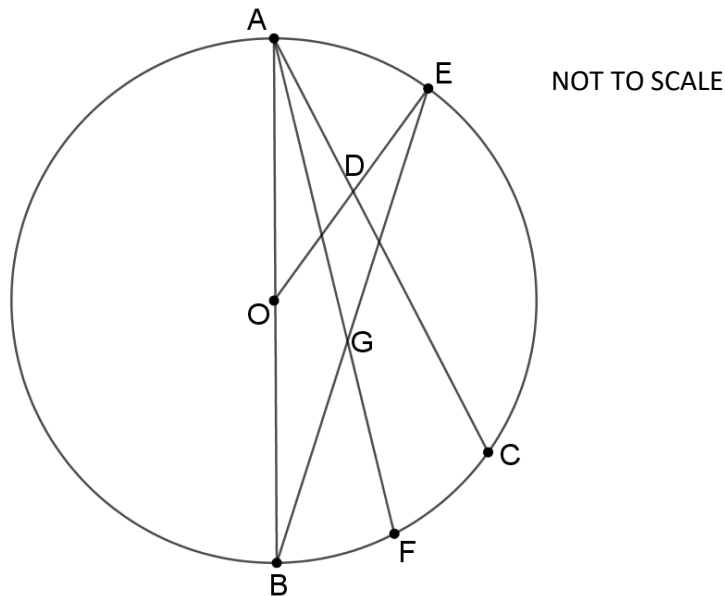
b) Find the coordinates of the vertex, focus and equation of the directrix of the parabola $x^2 + 6x + 8y - 7 = 0$ 4

c) Write down the Cartesian equation of the locus of a point $P(x, y)$ where $x = 2 \cos \theta$ and $y = \frac{1}{2} \sin \theta$ 2



In the diagram Q is the point $(-1, 0)$, R is the point $(1, 0)$, and P is another point on the circle with centre O and radius 1. Let $\angle POR = \alpha$ and $\angle PQR = \beta$, and let $\tan \beta = m$.

- a) Explain why $\triangle OPQ$ is isosceles, and hence deduce that $\alpha = 2\beta$ 2
- b) Find the equation of the line PQ in terms of m 2
- c) Show that the x -coordinates of P and Q are solutions of the equation $(1 + m^2)x^2 + 2m^2x + m^2 - 1 = 0$ 2
- d) Write an expression for the sum of the roots of this quadratic equation 1
- e) Hence or otherwise find the coordinates of P in terms of m 2

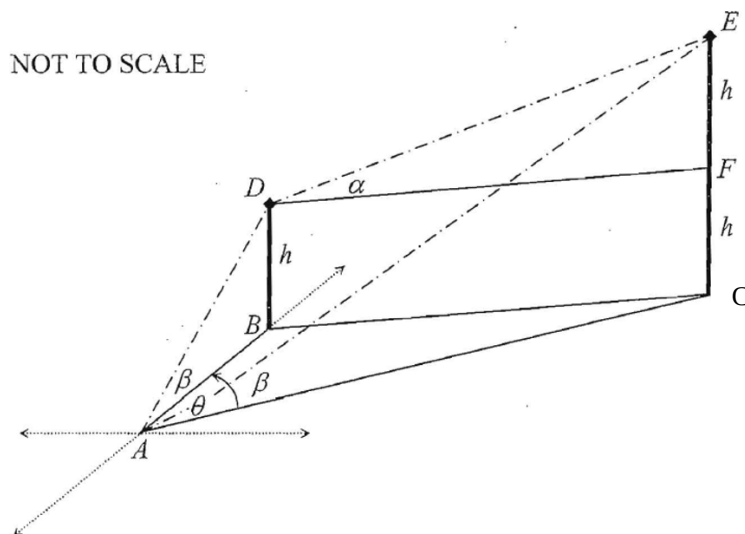


- a) In the figure, AOB is the diameter of a circle centre O . D is a point on chord AC such that $DA = DO$ and OD is produced to E on the circumference. AF is the bisector of $\angle BAC$ and cuts BE at G .

Draw the diagram into your answer booklet

Prove that

- | | |
|--|---|
| i) $GA = GB$ | 3 |
| ii) $AOGE$ is a cyclic quadrilateral | 2 |
| iii) If CE is joined, then $CE \parallel AF$ | 1 |
- c) A man standing on level ground at A notices two vertical towers BD and CE . The foot of tower BD , B , is due North of A and the foot of tower CE , C , is on a bearing of θ from A . The height of BD is h metres and the height of CE is twice the height of BD . The angle of elevation from A to the top of both towers is β . The angle of elevation to the top of CE from the top of BD is α



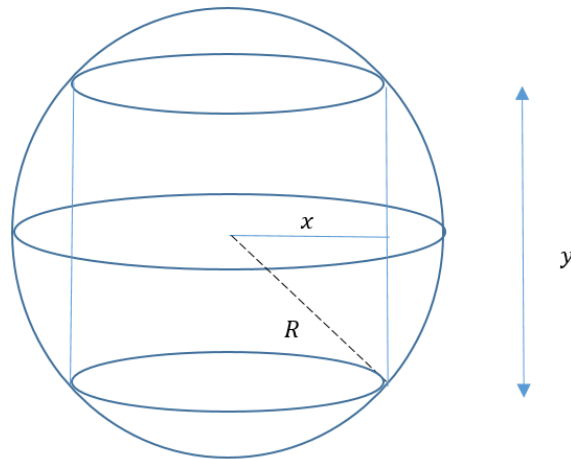
- | | |
|---|---|
| i) Show that $AC = 2h \cot \beta$ | 1 |
| ii) Find similar expressions for AB and BC | 2 |
| iii) Using the cosine rule, or otherwise, show that | 2 |

$$\cos \theta = \frac{5 \cot^2 \beta - \cot^2 \alpha}{4 \cot^2 \beta}$$

Question 8 (8 marks)

Marks

- a) A right cylinder of radius x and height y is inscribed in a sphere of fixed radius R .



(The volume of a sphere is $V = \frac{4}{3}\pi r^3$ and the volume of a right cylinder is $V = \pi r^2 h$)

- i) Show that the volume of the cylinder V , in the diagram above can be written as

2

$$V = \pi R^2 y - \frac{\pi y^3}{4}$$

- ii) Prove that the maximum volume of the cylinder occurs when $y = \frac{2R}{\sqrt{3}}$

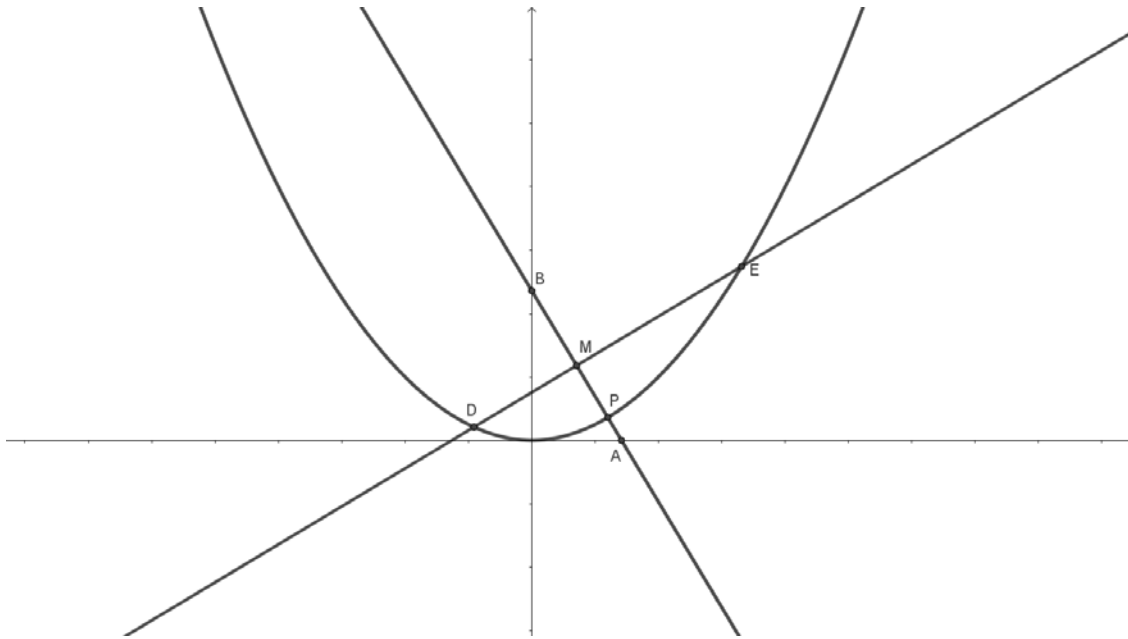
3

- iii) Find the maximum volume of the cylinder V in terms of R in simplest form

2

- iv) Show that when the cylinder has maximum volume, the ratio of the volume of the cylinder to the volume of the sphere is $1 : \sqrt{3}$.

1



The point $P(2at, at^2)$ lies on the parabola $4ay = x^2$.

The equation of the normal to the parabola at the point P is given by

$$y = -\frac{1}{t}x + 2a + at^2$$

- a) Find the coordinates of the points **A** and **B**, the x and y intercepts of the normal and hence find the coordinates of their midpoint **M** 3
- b) Find the equation of the normal to the normal in part a) that passes through the midpoint **M** 2
- c) The normal in part c) cuts the parabola at **D** and **E**. Without finding their coordinates, find the locus of their midpoint, stating all restrictions. 2

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Question 1 (18 marks)

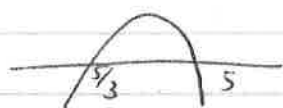
2 a) $2x^3 - 128$
 $= 2(x^3 - 64)$
 $= 2(x-4)(x^2 + 4x + 16)$

3 b) $\frac{2x}{5-x} \geq 1$

$$2x(5-x) - (5-x)^2 \geq 0$$

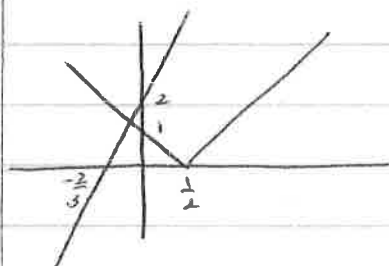
$$(5-x)(2x - (5-x)) \geq 0$$

$$(5-x)(3x-5) \geq 0$$



$$\frac{5}{3} \leq x \leq 5$$

3 c) $|2x-1| < 3x+2$



$$-(2x-1) = 3x+2$$

$$-1 = 5x$$

$$x = -\frac{1}{5}$$

$$\therefore x > -\frac{1}{5}$$

3 d) $x^{2/3} + x^{1/3} - 6 = 0$

Let $u = x^{1/3}$

$$u^2 + u - 6 = 0$$

$$(u+3)(u-2) = 0$$

$$u = -3$$

$$u = 2$$

$$x^{1/3} = -3$$

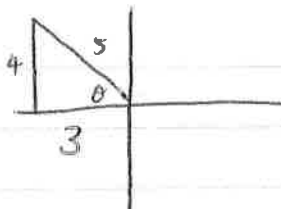
$$x^{1/3} = 2$$

$$x = (-3)^3 \quad x = 2^3$$

$$x = -27 \text{ or } +8$$

2

e)



$$\begin{aligned}\sin 2\theta &= 2 \sin \theta \cos \theta \\ &= 2 \times \frac{4}{5} \times -\frac{3}{5} \\ &= -\frac{24}{25}\end{aligned}$$

2

f)

$$\begin{aligned}&\frac{\cos(270 - \theta)}{\cos(180 - \theta)} \\ &= \frac{\cos(180 + (90 - \theta))}{\cos(180 - \theta)} \\ &= \frac{-\cos(90 - \theta)}{-\cos \theta} \\ &= \frac{-\sin \theta}{-\cos \theta} \\ &= \tan \theta\end{aligned}$$

3

g)

$$\begin{aligned}2 \tan^3 \theta - 2 \tan^2 \theta - 2 \tan \theta + 3 &= 0 \\ \tan^2 \theta (2 \tan \theta - 3) - (2 \tan \theta - 3) &= 0 \\ (2 \tan \theta - 3)(\tan^2 \theta - 1) &= 0 \\ \tan^2 \theta = \pm 1 &\quad \tan \theta = \frac{3}{2} \\ \theta = 45, 135, 225, 315, 56^\circ 19', 236^\circ 19'\end{aligned}$$

Question 2 (20 marks)

4 a) LHS = $\sin(x + \frac{\pi}{4})$
 $= \sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4}$
 $= \frac{\sin x}{\sqrt{2}} + \frac{\cos x}{\sqrt{2}}$
 $= \frac{\sin x + \cos x}{\sqrt{2}}$
 $= \text{RHS}$

$$\sin(x + \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$$

$$x + \frac{\pi}{4} = 60^\circ, 120^\circ$$

$$x + 45 = 60^\circ, 120^\circ$$

$$x = 15^\circ, 75^\circ$$

3 b) LHS = $\frac{\cos 2\theta}{\sin \theta} + \frac{\sin 2\theta}{\cos \theta}$ OR LHS
 $= \frac{1 - 2\sin^2 \theta}{\sin \theta} + \frac{2\sin \theta \cos \theta}{\cos \theta}$ $= \frac{\cos 2\theta \cos \theta + \sin 2\theta \sin \theta}{\sin \theta \cos \theta}$
 $= \frac{1 - 2\sin^2 \theta}{\sin \theta} + \frac{2\sin^2 \theta}{\sin \theta}$ $= \frac{\cos(2\theta - \theta)}{\sin \theta \cos \theta}$
 $= \frac{1}{\sin \theta}$ $= \frac{\cos \theta}{\sin \theta \cos \theta}$
 $= \operatorname{cosec} \theta$ $= \operatorname{cosec} \theta$
 $= \text{RHS}$ $= \text{RHS}$

2 c) $m_1 = 2, m_2 = 4$
 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $= \left| \frac{2 - 4}{1 + 2 \times 4} \right|$
 $= \frac{2}{9}$
 $\theta = 12^\circ 32' \text{ (or } 12.53^\circ \text{ or } 0.219 \text{ radians)}$

2 d) $A(-1, 4)$ $B(x, y)$

$-3 : 2$

$$8 = \frac{-3x - 2}{-3 + 2}$$

$$-8 = -3x - 2$$

$$3x = 6$$

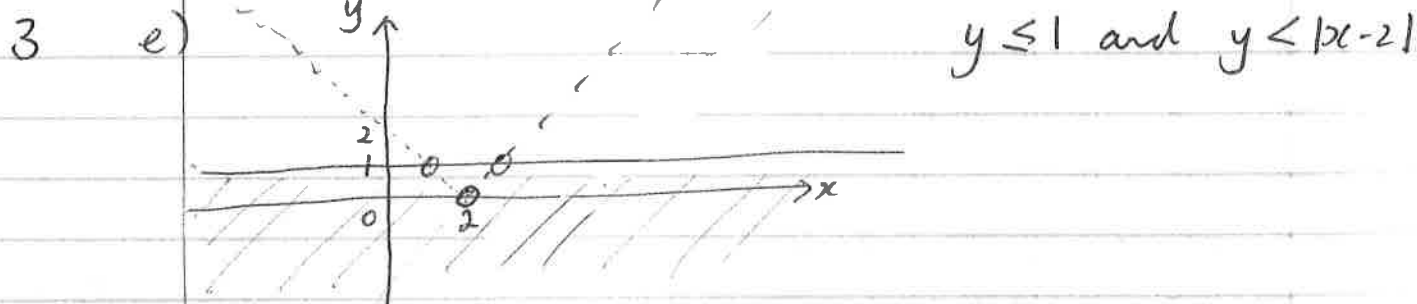
$$x = 2$$

$$-5 = \frac{-3y + 2(4)}{-3 + 2}$$

$$5 = -3y + 8$$

$$3y = 3$$

$$y = 1$$



2 f)

$$\frac{2^{1001} + 2^{1000}}{3}$$

$$= \frac{2^{1000} (2 + 1)}{3}$$

$$= 2^{1000}$$

4 g) i) $f(x) = a(x - b)(x + c)^d$

$b = -1$

$c = -3$

$d = 3$

$f(x) = a(x + 1)(x - 3)^3$

$(1, -16)$

$-16 = a(2)(-2)^3$

$\therefore a = 1$

ii) $D: -1 \leq x \leq 0$ or $x \geq 3$

Question 3 (13 marks)

a)

$$(i) (1-x)^7 \cdot 1 + x \cdot 7(1-x)^6 \cdot (-1)$$

$$= (1-x)^7 - 7x(1-x)^6$$

$$= (1-x)^6 (1-x-7x)$$

$$= (1-x)^6 (1-8x)$$

$$(ii) y = x \cdot \sqrt[3]{x}$$

$$y = x \times x^{1/3}$$

$$y = x^{4/3}$$

$$\frac{dy}{dx} = \frac{4}{3} x^{1/3}$$

$$= \frac{4}{3} \sqrt[3]{x}$$

$$(iii) \frac{dy}{dx} = \frac{(2+x) \cdot (-2x) - (5-x^2) \cdot 1}{(2+x)^2}$$

$$= \frac{-4x - 2x^2 - 5 + x^2}{(2+x)^2}$$

$$= \frac{-4x - 5 - x^2}{(2+x)^2}$$

$$b) y = 3x^2 - 2ax + 1$$

$$\frac{dy}{dx} = 6x - 2a$$

$$\text{When } x=2, \frac{dy}{dx} = -1$$

$$-1 = 12 - 2a$$

$$2a = 13$$

$$a = \frac{13}{2}$$

$$(c) \ y = 4x - x^3$$

$$\frac{dy}{dx} = 4 - 3x^2$$

$$\frac{d^2y}{dx^2} = -6x$$

concave up when

$$\frac{d^2y}{dx^2} > 0$$

$$-6x > 0$$

$$x < 0$$

$$(d) \ y = 3x^2 - 2x$$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3(x+h)^2 - 2(x+h) - (3x^2 - 2x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 - 2x - 2h - 3x^2 + 2x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 2h}{h}$$

$$= \lim_{h \rightarrow 0} 6x + 3h - 2$$

$$= 6x - 2$$

3.

Question 4 (11 marks)

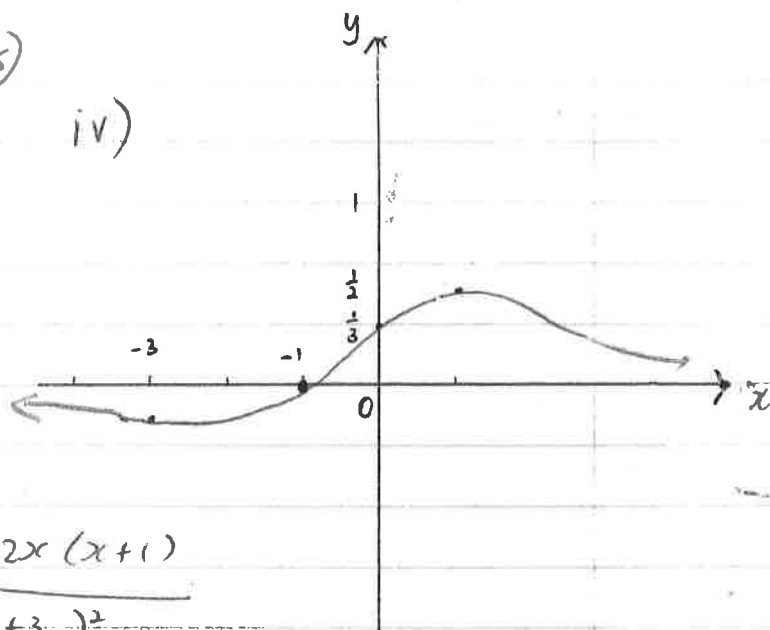
6 a) $f(x) = \frac{x+1}{x^2+3}$ (iv)

i) x intercept: $y = 0$

$x = -1$

y-intercept: $x = 0$

$y = \frac{1}{3}$



ii) $f'(x) = \frac{x^2+3 - 2x(x+1)}{(x^2+3)^2}$

$= \frac{x^2+3 - 2x^2-2x}{(x^2+3)^2}$

$= \frac{-x^2-2x+3}{(x^2+3)^2}$

Stat pts: $f'(x) = 0$

$-x^2-2x+3 = 0$

$-(x^2+2x-3) = 0$

$-(x+3)(x-1) = 0$

$x = -3, y = \frac{-2}{12} = -\frac{1}{6}$

$x = 1, y = \frac{2}{4} = \frac{1}{2}$

x	-4	-3	0	1	2
$f'(x)$	$-\frac{5}{49}$	0	$\frac{3}{9}$	0	$-\frac{5}{49}$
		\	/	\	

$\therefore$ min t.p. at $(-3, -\frac{1}{6})$ and max t.p. at $(1, \frac{1}{2})$

iii) $\lim_{x \rightarrow \infty} \frac{x+1}{x^2+3} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x} + \frac{1}{x^2}}{1 + \frac{3}{x^2}}$

$= \lim_{x \rightarrow \infty} \frac{\frac{x}{x^2} + \frac{1}{x^2}}{\frac{x^2}{x^2} + \frac{3}{x^2}} = 0$

$$5 \quad b) \quad 4x^2 - 2x - 1 = 0$$

$$i) \quad \frac{1}{2\alpha} + \frac{1}{2\beta}$$

$$= \frac{\beta + \alpha}{2\alpha\beta}$$

$$= \frac{\frac{2}{4}}$$

$$2 \left(-\frac{1}{4}\right)$$

$$= -1$$

$$ii) \quad \alpha^2 + \beta^2$$

$$= (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \left(\frac{2}{4}\right)^2 - 2\left(-\frac{1}{4}\right)$$

$$= \frac{1}{4} + \frac{1}{2}$$

$$= \frac{3}{4}$$

$$iii) \quad |\alpha - \beta| = \sqrt{(\alpha - \beta)^2}$$

$$= \sqrt{\alpha^2 + \beta^2 - 2\alpha\beta}$$

$$= \sqrt{\frac{3}{4} - 2\left(-\frac{1}{4}\right)}$$

$$= \frac{\sqrt{5}}{2}$$

Question 5 (9 marks)

3 a) $(k+1)x^2 - 2(k-1)x + (2k-5)$

Perfect square : $\Delta = 0$

$$(-2(k-1))^2 - 4(k+1)(2k-5) = 0$$

$$4k^2 - 8k + 4 - 4(2k^2 - 3k - 5) = 0$$

$$4k^2 - 8k + 4 - 8k^2 + 12k + 20 = 0$$

$$-4k^2 + 4k + 24 = 0$$

$$k^2 - k - 6 = 0$$

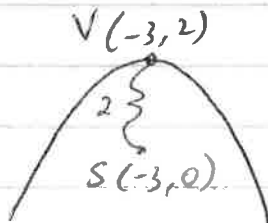
$$(k-3)(k+2) = 0$$

$$k = 3 \text{ or } -2$$

4 b) $x^2 + 6x + 8y - 7 = 0$

$$(x+3)^2 = -8y + 16$$

$$y = 4 - \dots = -8(y-2)$$



$$4a = 8$$

$$a = 2$$

Vertex : $(-3, 2)$

Focus : $S(-3, 0)$

Directrix : $y = 4$

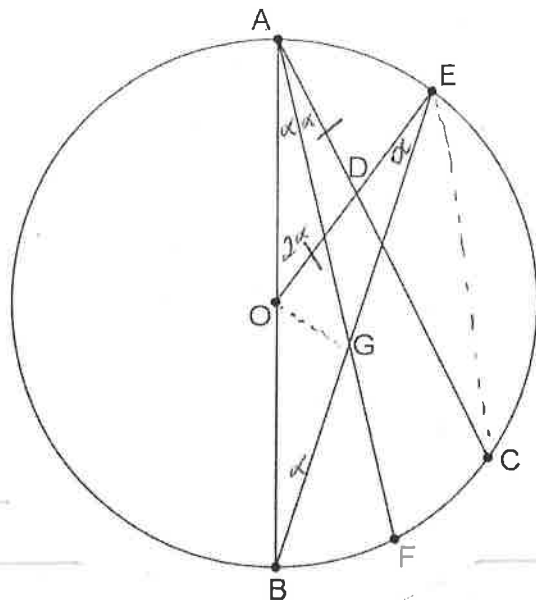
2 c) $x = 2 \cos \theta$ $y = \frac{1}{2} \sin \theta$

$$\cos \theta = \frac{x}{2} \quad (1) \quad \sin \theta = 2y \quad (2)$$

$$(1)^2 + (2)^2 \quad \cos^2 \theta + \sin^2 \theta = \left(\frac{x}{2}\right)^2 + (2y)^2$$

$$\frac{x^2}{4} + 4y^2 = 1$$

Question 6 (9 marks)



- 2 a) $OQ = OP$ (equal radii of circle)
 $\therefore \triangle OQP$ is isosceles (2 equal sides)
 $\therefore \angle OQP = \angle OPQ = \beta$ (base angles of isosceles triangle)
 $\therefore \alpha = 2\beta$ (exterior angle of a triangle)

- 2 b) $\tan \beta = m \quad (-1, 0)$
 $y - 0 = m(x + 1)$

Equation of PQ: $y = mx + m$

- 2 c) $x^2 + y^2 = 1$ (1)
 $y = mx + m$ (2)

Solving simultaneously gives points of intersection P, Q.

Sub (1) into (2)

$$x^2 + (mx + m)^2 = 1$$

$$x^2 + m^2x^2 + 2m^2x + m^2 - 1 = 0$$

1 d) $\alpha + \beta = \frac{-2m^2}{1+m^2}$

2 e) One of the roots is -1

$$\therefore -1 + \beta = \frac{-2m^2}{1+m^2}$$

$$\beta = \frac{-2m^2}{1+m^2} + 1$$

$$= \frac{-2m^2 + 1 + m^2}{1+m^2}$$

$$= \frac{1-m^2}{1+m^2}$$

$$\therefore P \left(\frac{1-m^2}{1+m^2}, \frac{2m}{1+m^2} \right)$$

Question 7

3 a) i) Let $\angle BAF = \alpha$

$\angle BAF = \angle FAC = \alpha$ (AF is the bisector of $\angle BAC$)

$\angle BAD = 2\alpha$ (given)

$OA = OD$ (given)

$\angle BAD = \angle AOD = 2\alpha$ (Equal angles opposite equal sides)

$2\angle ABE = \angle AOD$ (Angle at centre is twice angle at circumference)

$\therefore \angle ABE = \alpha$

Now $\angle ABE = \angle GAO = \alpha$

$\therefore GA = GB$ (Equal sides opposite equal angles)

2 ii) $OB = OE$ (Equal radii)

$\angle ABE = \angle OEB = \alpha$ (Equal angles opposite equal sides)

$\therefore \angle BAF = \angle OEB = \alpha$

$\therefore AOG E$ is a cyclic quadrilateral

(Intersect OE subtends equal angles at A and E on the same side of the intersect)

1 iii) $\angle ABE = \angle ACE = \alpha$ (Angles in the same segment)

$\therefore \angle FAC = \angle ACE = \alpha$

$\therefore CE \parallel AF$ (Alternate angles are equal)

1 (b) i) In $\triangle ACE$

$$\tan(90-\beta) = \frac{AC}{2h}$$

$$AC = 2h \tan(90-\beta) \\ = 2h \cot \beta$$

2 ii) In $\triangle ABD$,

$$\tan(90-\beta) = \frac{AB}{h}$$

$$AB = h \tan(90-\beta) \\ = h \cot \beta$$

$$\tan(90-\alpha) = \frac{DF}{h}$$

$$DF = h \cot \alpha$$

$$\text{But } DF = BC$$

$$\therefore BC = h \cot \alpha$$

2 iii)

$$\cos \theta = \frac{AB^2 + AC^2 - BC^2}{2 \times AB \times AC}$$

$$= \frac{h^2 \cot^2 \beta + 4h^2 \cot^2 \beta - h^2 \cot^2 \alpha}{2 \times h \cot \beta \times 2h \cot \beta}$$

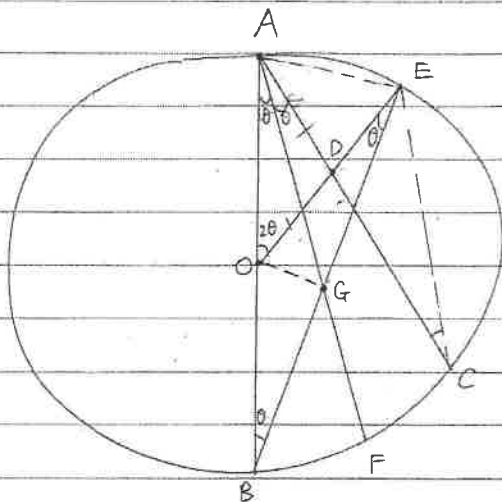
$$= \frac{h^2 (\cot^2 \beta + 4 \cot^2 \beta - \cot^2 \alpha)}{4h^2 \cot^2 \beta}$$

$$= \frac{5 \cot^2 \beta - \cot^2 \alpha}{4 \cot^2 \beta}$$

Model Response

Question 7:

a)

i) RTP: $GA = GB$ $\therefore$ AF bisects $\angle BAC$ $\therefore \angle BAF = \angle FAC$ Let $\angle BAF = \angle FAC = \theta$
 $\therefore \angle OAD = 2\theta$
 In triangle ODA
 $\therefore AD = DO$ (Given)

 $\therefore$ Triangle ODA is isosceles (Two sides of an isosceles triangle are equal)

 $\therefore \angle OAD = \angle ODA$ (Base angles of isosceles triangles are equal)
 $\therefore \angle OAD = \angle ODA = 2\theta$ $\therefore \angle AOE = 2\theta$
 $\therefore \angle ABE = \frac{2\theta}{2} = \theta$ (Angle at circumference standing on the same arc, is half the angle at centre on the same arc)
 $\therefore \angle ABE = \theta$ $\therefore \angle ABE = \angle BAF = \angle FAC = \theta$ $\therefore \angle BAF = \theta = \angle ABE$
 $\therefore$ Triangle AGB is isosceles (Two ~~sides~~^{angle} are equal in an isosceles triangle)

 $\therefore AG = BG$ (Base sides of an isosceles triangle are equal)

ii). Construct line AE and OG

In triangle BOE,

 $OB = OE$ (Radii in the same circle are equal)
 $\therefore \triangle BOE$ is isosceles (Two sides of an isosceles triangle are equal)

 $\therefore \angle OBE = \angle OEB$ (Base angles of isosceles triangles are equal)
 $\therefore \angle OBE = \theta$ $\therefore \angle OEB = \theta$

In quadrilateral, AOGE

 because OG subtends equal angles of θ at $\angle OAG$ and $\angle OEG$, Therefore it is cyclic.
iii). $\therefore \angle AOE = 2\theta$
 $\therefore \angle ACE = \theta$ (Angle at circumference standing on the same arc is half the angle at centre standing on the same arc)
 $\therefore \angle ACE = \theta = \angle FAC$
 $\therefore AF \parallel EC$ (Alternate angles $\angle FAC$ and $\angle ACE$ are only equal if $AF \parallel EC$)

Additional writing space on back page.

Question 8

2

$$\begin{aligned} \text{i) } V &= \pi r^2 h \\ &= \pi x^2 h \\ &= \pi x^2 y \end{aligned}$$

$$R^2 = x^2 + \left(\frac{y}{2}\right)^2$$

$$x^2 = R^2 - \frac{y^2}{4}$$

$$= \pi \left(R^2 - \frac{y^2}{4} \right) y$$

$$= \pi R^2 y - \frac{\pi y^3}{4}$$

2

3

$$\text{ii) } \frac{dV}{dy} = \pi R^2 - \frac{3\pi y^2}{4}$$

$$\frac{d^2V}{dy^2} = -\frac{6\pi y}{4} = -\frac{3\pi y}{2}$$

$$\text{Let } \frac{dV}{dy} = 0$$

$$\pi R^2 - \frac{3\pi y^2}{4} = 0$$

$$\frac{3\pi y^2}{4} = \pi R^2$$

$$y^2 = \frac{\pi R^2 \times 4}{3\pi}$$

$$y = \pm \sqrt{\frac{4R^2}{3}}$$

$$y = \frac{2R}{\sqrt{3}} \quad y > 0 \text{ as it is a length}$$

(3)

- 1 - first der
- 1 - ^{show} find y
- 1 - Test

$$\text{Test } y = \frac{2R}{\sqrt{3}}$$

$$\frac{d^2V}{dy^2} = -\frac{3\pi}{2} \left(\frac{2R}{\sqrt{3}} \right) < 0$$

for all $R > 0$. $\therefore$ max at $y = \frac{2R}{\sqrt{3}}$

2

$$\text{iii) Max } V = \pi R^2 \left(\frac{2R}{\sqrt{3}} \right) - \frac{\pi}{4} \left(\frac{2R}{\sqrt{3}} \right)^3$$

$$= \frac{2\pi R^3}{\sqrt{3}} - \frac{2^3 \pi R^3}{4 \times 3\sqrt{3}}$$

$$= \frac{6\pi R^3 - 2\pi R^3}{3\sqrt{3}}$$

$$= \frac{4\pi R^3}{3\sqrt{3}}$$

Sub in - mark
Correct answer - 1

(2)

1

$$\text{iv) } V_{\text{SPHERE}} = \frac{4\pi R^3}{3}$$

$$V_{\text{CYL}} = \frac{4\pi R^3}{3\sqrt{3}}$$

$$V_{\text{SP}} : V_{\text{CYL}}$$

$$\frac{\frac{4\pi R^3}{3}}{\frac{4\pi R^3}{3\sqrt{3}}}$$

$$\frac{1}{3} \div \frac{1}{3\sqrt{3}}$$

$$\sqrt{3} \div 1$$

$$\therefore V_{\text{CYL}} : V_{\text{SP}} = 1 : \sqrt{3}$$

(1)

Must show

Question 9

$$4ay = x^2 \quad P(2at, at^2)$$

3 a) $y' = \frac{2x}{4a} = \frac{x}{2a} = \frac{2at}{2a} = t$

$$m_{\text{normal}} = -\frac{1}{t}$$

$$y - at^2 = -\frac{1}{t}(x - 2at) \quad \text{Eq. of normal at } P$$

$$\boxed{y = -\frac{1}{t}x + 2a + at^2} \quad (1)$$

$$x\text{-intercept: } y = 0 \rightarrow x = (2a + at^2)t$$

$$y\text{-intercept: } x = 0 \rightarrow y = 2a + at^2$$

$$B(0, 2a + at^2), \quad A(2at + at^3, 0)$$

$$M \text{ midpoint of } AB \left(at + \frac{1}{2}at^3, a + \frac{a}{2}t^2 \right)$$

2 b) Normal to (1) at M

$$y - \left(a + \frac{a}{2}t^2 \right) = t \left(x - at - \frac{1}{2}at^3 \right)$$

$$y = tx - at^2 - \frac{1}{2}at^4 + a + \frac{a}{2}t^2$$

$$\boxed{y = tx - \frac{1}{2}at^4 - \frac{a}{2}t^2 + a}$$

2 c) $4ay = 4atx - 2a^2t^4 - 2a^2t^2 + 4a^2$

$$x^2 - 4atx + 2a^2(t^4 + t^2 - 2) = 0$$

$$\text{midpoint} \left(2at, -\frac{1}{2}at^4 + \frac{3}{2}at^2 + a \right)$$

$$y = \frac{-1}{32a^3}x^4 + \frac{3}{8a}x^2 + a$$

$$\begin{aligned}
 \Delta &= 16a^2t^2 - 8a^2(t^4 + t^2 - 2) \\
 &= -8a^2t^4 + 8a^2t^2 + 16a^2 \\
 &= 8a^2(-t^4 + t^2 + 2) \\
 &= 8a^2(-t^2 + 2)(t^2 + 1) \geq 0
 \end{aligned}$$

$$-t^2 + 2 \geq 0$$

$$-\sqrt{2} \leq t \leq \sqrt{2}$$

$$-2a\sqrt{2} \leq x \leq 2a\sqrt{2}$$